Arista Access Node QoS Congestion Event Resolution SOP for MALL-ACC-21

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2918

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for addressing QoS congestion events on the Arista Access Node MALL-ACC-21, specifically in the context of microbursts, interface tail drops, and QoS queue starvation. The Arista access node is responsible for aggregating multiple access links and forwarding traffic to the core network. In this scenario, the access node is running in a high-availability setup with dual links (active-active), and the QoS configuration is designed to prioritize critical traffic flows. However, in certain situations, the access node may experience congestion due to excessive microbursts or uneven traffic distribution, leading to interface tail drops and QoS queue starvation. This SOP aims to identify and resolve the root cause of QoS congestion events, thereby ensuring optimal network performance and minimizing the likelihood of service disruptions.

Pre-requisite Checks

- Verify that the Malleswaram Node is accessible and reachable via remote hands or on-site personnel.

- Check the status of redundant links (if any) and ensure they are functioning correctly.

- Review the current QoS configuration and verify that it aligns with the organization's service level agreements (SLAs).

Potential Root Causes

Cause	Probability
Microbursts due to misconfigured or malfunctioning network devices	High
Interface tail drops caused by excessive packet buffering	Medium
QoS queue starvation due to inadequate queue sizing or priority settings	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status: UP/DOWN, Speed: 10G, Duplex: FULL, Description: This command is used to verify the link status and speed of the interface(s) involved in the QoS congestion event.

Command: show qos queue statistics, Expected Output: Queue statistics: packets dropped, bytes dropped, packet loss percentage, Description: This command provides detailed statistics on QoS queue performance, including packets dropped, bytes dropped, and packet loss percentage.

Command: show qos policy, Expected Output: QoS policy details: priority, rate, burst size, and queue size, Description: This command
```

```
displays the current QoS policy configuration, including priority,
rate, burst size, and queue size settings.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy settings, Command: configure qos policy priority rate burst-size queue-size , Description: This step involves reviewing and adjusting the QoS policy settings to ensure they are aligned with the organization's SLAs and service requirements.

Step 2: Step: Adjust interface buffer sizes, Command: configure interface buffer-size , Description: This step involves adjusting the interface buffer sizes to prevent excessive packet buffering and interface tail drops.

Step 3: Step: Disable or adjust microburst mitigation, Command: configure qos policy microburst-mitigation , Description: This step involves disabling or adjusting microburst mitigation settings to prevent excessive packet drops and queue starvation.

Post-Check Verification

```
Command: show qos queue statistics, Expected Output: Queue statistics:
packets dropped, bytes dropped, packet loss percentage, Description:
This command is used to verify the effectiveness of the resolution
steps and ensure that QoS congestion events are resolved.

Command: show qos policy, Expected Output: QoS policy details:
priority, rate, burst size, and queue size, Description: This command
is used to verify that the QoS policy settings are correctly
configured and aligned with the organization's SLAs.
```

Roll-back Procedures

- Step: Restore original QoS policy settings, Command: configure qos policy priority rate burst-size queue-size , Description: This step involves restoring the original QoS policy settings in case the resolution steps fail to resolve the QoS congestion event.

- Step: Restore original interface buffer sizes, Command: configure interface buffer-size , Description: This step involves restoring the original interface buffer sizes in case the resolution steps fail to resolve the QoS congestion event.